



# VIGNAN'S

FOUNDATION FOR SCIENCE, TECHNOLOGY & RESEARCH

(Deemed to be University) - Estd. u/s 3 of UGC Act 1956

## Module Bank

Module 2

Academic Year 2025-26

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Staff Name: Dr. Sreekar Guddeti

Program Name: B. Tech

Branch: AI/ML

Year: 1

Semester: 1

Course: EP

Code: 25PY101

Section number: 28, 34

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### Instructions:

1. Answer all questions.
2. BT stands for Blooms Taxonomy.

# PART-A

## 1. Quantum Free Electron Theory (QFET)

[10 M]

- a) Compare and contrast the postulates of classical free electron theory with that of quantum free electron theory. [BT 1][6 M]
- b) What are the merits of quantum free electron theory? [BT 1][2 M]
- c) What are the demerits of quantum free electron theory? [BT 1][2 M]

## 2. Fermi-Dirac Distribution

[10 M]

- a) Plot the probability of occupancy of energy level in the case of Maxwell-Boltzmann distribution and Fermi-Dirac distribution at
  - i.  $T = 0K$ ,
  - ii.  $T \neq 0K$ , and
  - iii.  $T = \infty K$  [BT 2][2 M]
- b) Five free electrons exist in a three-dimensional infinite potential well with all three widths equal to  $a = 12 \text{ \AA}$ .
  - i. Determine the Fermi energy level at  $T = 0K$ .
  - ii. Repeat part (i) for 13 electrons. [BT 3][4 M]
- c) Fermi level of a metal is  $6.25 \text{ eV}$  and electrons in this material follow the Fermi-Dirac distribution. Calculate the temperature at which there is 1% probability that a state  $0.30 \text{ eV}$  below the Fermi energy level will **not** contain an electron. [BT 3][4 M]

## 3. Electronic specific heat of solids

[10 M]

- a) Define specific heat of a material. What is its SI unit? [BT 1][2 M]
- b) Specific heat of mercury ( ${}_{80}^{200}\text{Hg}$ ) is  $0.14 \text{ J g}^{-1}\text{K}^{-1}$ . Report its specific heat in SI units. [BT 1][2 M]
- c) Derive the expression for specific heat capacity from classical free electron theory and quantum free electron theory. [BT 4][6 M]

## 4. Density of states

[10 M]

- a) What is the meaning of density of states function? What is its SI unit? [BT 1][2 M]
- b) If the density of states of a material is a constant  $Z(E) = k$  and the electron density is  $n$ ,
  - i. what is the Fermi level at  $T = 0K$ ?
  - ii. What is the Fermi level at  $T \neq 0K$ ? [BT 3][4 M]
- c) Sketch the density of states function
  - i. for the conduction band of an intrinsic semiconductor.
  - ii. Repeat part (i) for the valence band. [BT 2][4 M]

## 5. E-k diagram

[10 M]

- a) Sketch the  $E$  vs  $k$  diagram for a free electron. [BT 1][1 M]
- b) Sketch the  $E$  vs  $k$  diagram for an electron in a periodic potential. [BT 1][1 M]

- c) Consider an intrinsic semiconductor with all electrons occupying states in the lowest valence band at  $T = 0K$ . Plot the  $E$  vs  $k$  showing occupancy of levels at both  $T = 0K$  and  $T \neq 0K$ . [BT 2][2 M]
- d) Two possible conduction bands are shown in the  $E$  vs  $k$  diagram given in Figure. 1(a). State which band will result in the heavier electron effective mass; state why [BT 2][2 M]
- e) Figure. 1(b) shows the parabolic  $E$  vs  $k$  relationship in the valence band for a hole in two particular semiconductor materials. Determine the effective mass (in units of the free electron mass) of the two holes. [BT 3][4 M]

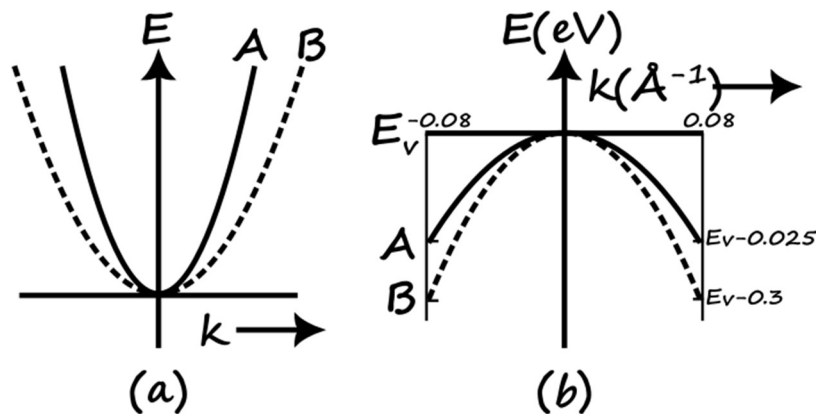


Figure 1: (a) Conduction and (b) valence bands of materials A and B.

## 6. Fermi level in semiconductors

[10 M]

- a) Sketch the energy band diagram showing the density of states, Fermi-Dirac probability function and areas representing electron and hole concentration for an intrinsic semiconductor at  $T \neq 0K$ . Assume electron effective mass is equal to hole effective mass. [BT 1, 2][3 M]
- b) Calculate position of Fermi level with respect to center of the bandgap in silicon for
- $T = 0 K$ , and
  - $T = 300 K$ .

The electron effective mass is  $1.08m_0$  and hole effective mass is  $0.56m_0$ . [BT 2][2 M]

- c) Draw, compare, and contrast the sketches of the energy band diagram for p-type semiconductor at  $T = 0K$  and  $T = 300K$ . [BT 3][2 M]
- d) Silicon atoms, at a concentration of  $7 \cdot 10^{15} cm^{-3}$ , are added to GaAs. Assume that the silicon atoms act as fully ionized dopant atoms and that 5% of the dopants replace Ga atoms and 95% replace As atoms.
- Determine the donor and acceptor concentrations.
  - Is the material n type or p type?

[BT 2][3 M]

## 7. p-n junction diode

[10 M]

- a) Write the equation and sketch the  $I - V$  characteristics of pn junction diode. [BT 1][2 M]
- b) Define reverse saturation current of pn junction diode. Derive it from the  $I - V$  characteristics. [BT 1][2 M]

- c) A silicon diode can be used to measure temperature by operating the diode at a fixed forward-bias current. The forward bias voltage is then a function of temperature. Assume that the reverse saturation current is proportional to  $\exp(-E_g/k_B T)$ . At  $T = 300K$ , the diode voltage is found to be  $0.60V$ . Determine the diode voltage at  $T = 310K$ . What is the percentage change in voltage? [BT 4][3 M]
- d) The pn junction diode can also be used as a temperature sensor by operating the diode at a fixed reverse-bias voltage. What is the percentage change in current for the same temperature rise? [BT 4][2 M]
- e) Which mode is more sensitive, the constant forward current mode or constant reverse voltage mode? [BT 2][1 M]

## 8. Solar cell

[10 M]

- a) Sketch the  $I - V$  characteristics of pn junction solar cell. [BT 1][1 M]
- b) Define short circuit current and open circuit voltage. [BT 1][2 M]
- c) A silicon solar cell under standard test conditions has short circuit current density of  $35 \text{ mA/cm}^2$ , and open circuit voltage of  $0.60 \text{ V}$ . The fill factor for the material is  $0.80$ . If the incident power is  $100 \text{ mW/cm}^2$ , calculate the efficiency of the cell. [BT 3][4 M]
- d) The energy band gap of a semiconductor is  $3 \text{ eV}$ . Is this material useful for solar cell applications? Give reasons. [BT 4][3 M]

## 9. Direct and indirect band gap semiconductors

[10 M]

- a) Sketch the  $E$  vs  $k$  diagram for
- a direct bandgap semiconductor, and
  - an indirect bandgap semiconductor. [BT 1][2 M]
- b) Why is direct band gap semiconductor useful for solar cell application? Crystalline silicon is indirect band gap semiconductor. What is the process employed in industry to “convert” it into a direct band gap semiconductor for solar cell application? [BT 2][2 M]
- c) Which material is used in blue LED? What is the bandgap of the material? [BT 3][3 M]
- d)  $\text{Al}_x\text{Ga}_{1-x}\text{As}$  is a ternary semiconductor and the energy band gap variation as a function of mole fraction  $x$  of  $\text{AlAs}$  is shown in Figure 1. Which composition can be used as material for blue LED application? [BT 4][3 M]

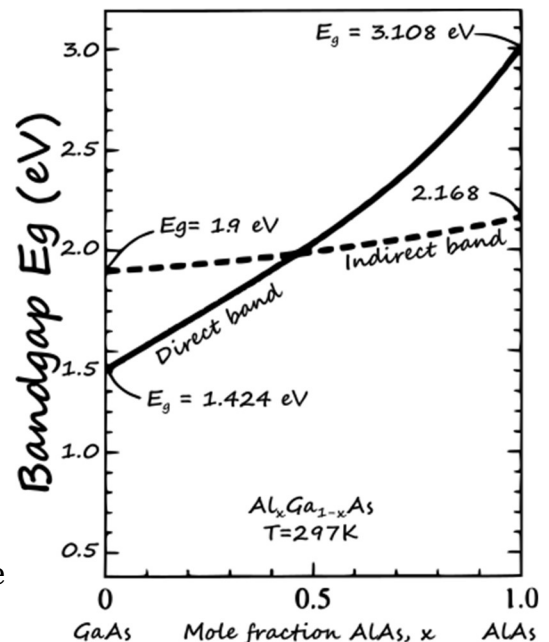


Figure 1. Energy Bandgap

**10. LED****[10 M]**

- a) What are the materials used for red, green, and blue LEDs? [BT 1][2 M]
- b) What are the mechanisms that decrease the efficiency of an LED? [BT 1][3 M]
- c) The energy band gap variation as a function of mole fraction  $x$  of  $GaAs_{1-x}P_x$  is shown in Figure 1. What is the range of colours that can be emitted if used for LED application? Report in wavelength. [BT 3][3 M]
- d) Calculate the critical angle at GaAs - air interface. The refractive index of GaAs is 3.8 at  $\lambda = 0.70 \mu m$ . [BT 3][2 M]

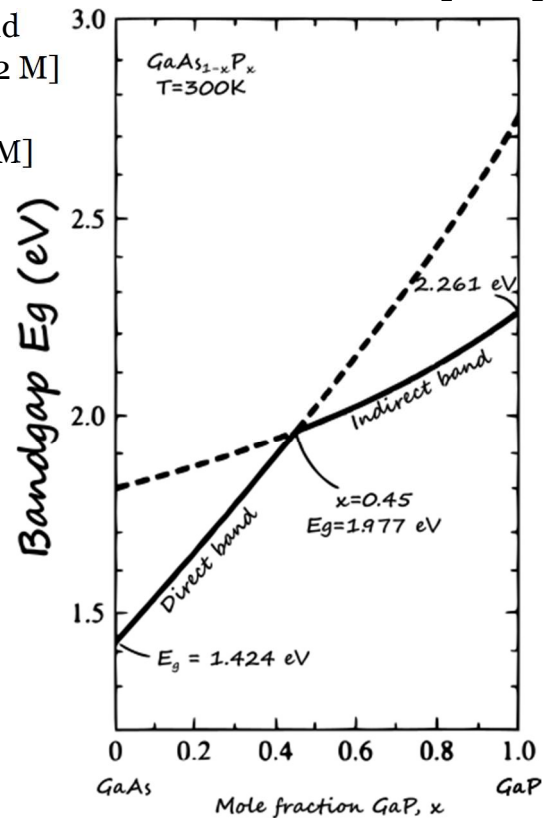


Figure 1. Energy band gap

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